Complete Results of *Parallelizing Node-Level Explainability in Graph Neural Networks*

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1. Complete results

This document presents the complete set of results for the paper *Parallelizing Node-Level Explainability in Graph Neural Networks*. These results were obtained for two models (GCN [1] and GAT [2]) using four explainability techniques (DeconvNet [3], GNNExplainer [4], Guided Backprop [5], and Saliency Maps [6]) on three citation network datasets: Cora, CiteSeer, and PubMed [7].

Table 1 serves as an index to all detailed result tables corresponding to each combination of explainability technique, dataset, and model architecture. For every combination, we evaluate multiple clustering configurations (ranging from 1 to 128 clusters) and vary the dropout rate between 0 and 1. Reported metrics include execution time, increases in nodes and edges, and the percentage of affected explanations and nodes.

Note that for the PubMed dataset, which is significantly larger than Cora and CiteSeer, experiments are limited to 16 clusters due to the GPU memory required to store the reconstructed dataset. Running larger cluster configurations on CPU would prevent a fair comparison with the GPU-based experiments on the smaller datasets, since CPU execution times would be

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disproportionately higher. Similarly, GNNExplainer is not run on PubMed because its memory requirements exceed the available GPU capacity.

Table 1: Summary of table references for each explainability technique evaluated on the Cora, CiteSeer, and PubMed datasets under GCN and GAT model architectures.

Explainability	**Cora**		**CiteSeer**		**PubMed**	
technique	GCN	GAT	GCN	GAT	GCN	GAT
DeconvNet	Table 2	Table 6	Table 10	Table 14	Table 18	Table 21
GNNExplainer	Table 3	Table 7	Table 11	Table 15	–	–
Guided Backprop	Table 4	Table 8	Table 12	Table 16	Table 19	Table 22
Saliency Map	Table 5	Table 9	Table 13	Table 17	Table 20	Table 23

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Table 2: Reconstruction with DeconvNet on Cora dataset using the GCN model.

Number of clusters	**Rate**	**Execution time (s)**	**Δnodes (%)**	**Δedges (%)**	**Affected explanations (%)**			**Affected nodes (%)**		
					0.2	0.5	0.7	0.2	0.5	0.7
1	1.00	17.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
16	1.00	3.07	720.35	774.97	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	2.61	576.00	498.22	10.78	1.14	0.37	0.46	0.03	0.01
	0.50	1.94	361.63	182.91	22.01	2.73	0.41	1.20	0.08	0.01
	0.20	1.58	218.28	60.21	27.66	4.21	1.33	1.64	0.12	0.04
	0.00	1.34	0.00	-13.93	31.09	5.83	1.44	2.04	0.17	0.04
32	1.00	2.10	1084.86	1149.77	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	1.75	871.05	729.23	14.99	2.10	0.48	0.70	0.06	0.01
	0.50	1.29	544.09	291.68	26.66	4.14	0.96	1.52	0.12	0.03
	0.20	1.00	323.38	96.23	32.61	5.32	1.18	2.09	0.16	0.03
	0.00	0.84	0.00	-18.57	37.92	7.05	1.92	2.63	0.21	0.05
64	1.00	1.61	1667.25	1736.23	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	1.41	1335.19	1109.95	17.36	1.70	0.26	0.77	0.05	0.01
	0.50	1.04	838.07	424.99	33.83	4.25	0.89	2.07	0.13	0.02
	0.20	0.89	498.71	139.39	39.81	6.39	1.44	2.77	0.19	0.04
	0.00	0.65	0.00	-24.40	45.46	8.42	2.10	3.57	0.26	0.06
128	1.00	1.64	2774.52	2860.34	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	1.44	2221.42	1827.06	34.42	9.49	2.81	1.83	0.28	0.08
	0.50	1.16	1385.89	710.21	56.91	20.57	8.38	4.80	0.69	0.24
	0.20	0.95	829.28	236.55	64.07	27.10	10.86	6.60	0.96	0.32
	0.00	0.78	0.00	-50.42	72.12	36.48	14.92	8.65	1.43	0.47

Table 3: Reconstruction with GNNExplainer on the Cora dataset using the GCN model.

Number of clusters	**Rate**	**Execution time (s)**	**Δnodes (%)**	**Δedges (%)**	**Affected explanations (%)**			**Affected nodes (%)**		
					0.2	0.5	0.7	0.2	0.5	0.7
1	0.00	1903.56	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
16	1.00	194.70	323.08	318.78	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	166.09	258.83	202.07	49.19	49.19	49.19	11.34	11.34	11.34
	0.50	141.33	166.10	81.02	61.48	61.48	61.48	25.38	25.38	25.38
	0.20	131.88	96.27	20.56	67.17	67.17	67.17	37.79	37.79	37.79
	0.00	122.96	0.00	-13.93	69.76	69.76	69.76	44.25	44.25	44.25
32	1.00	109.53	433.71	416.12	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	93.92	347.27	259.00	54.65	54.65	54.65	16.42	16.42	16.42
	0.50	74.87	217.43	102.90	68.17	68.17	68.17	39.31	39.31	39.31
	0.20	67.53	127.14	29.20	71.60	71.60	71.60	47.32	47.32	47.32
	0.00	61.39	0.00	-18.57	75.41	75.41	75.41	57.89	57.89	57.89
64	1.00	64.75	623.12	583.50	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	54.38	498.56	368.55	58.16	58.16	58.16	20.65	20.65	20.65
	0.50	40.01	307.61	133.72	72.08	72.08	72.08	46.52	46.52	46.52
	0.20	34.89	189.11	34.99	75.59	75.59	75.59	57.95	57.95	57.95
	0.00	30.37	0.00	-24.40	79.10	79.10	79.10	67.83	67.83	67.83
128	1.00	44.94	1006.94	937.12	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	36.32	806.20	601.42	69.79	69.79	69.79	26.14	26.14	26.14
	0.50	25.46	507.31	231.09	80.76	80.76	80.76	55.72	55.72	55.72
	0.20	19.43	302.22	67.09	84.23	84.23	84.23	68.83	68.83	68.83
	0.00	15.78	0.00	-50.42	86.45	86.45	86.45	83.99	83.99	83.99

Table 4: Reconstruction with Guided Backprop on the Cora dataset using the GCN model.

Number of clusters	**Rate**	**Execution time (s)**	**Δnodes (%)**	**Δedges (%)**	**Affected explanations (%)**			**Affected nodes (%)**		
					0.2	0.5	0.7	0.2	0.5	0.7
1	1.00	16.82	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
16	1.00	3.09	720.35	774.97	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	2.61	573.38	487.86	12.37	0.96	0.30	0.54	0.03	0.01
	0.50	1.93	361.26	191.27	21.57	2.95	0.63	1.17	0.08	0.02
	0.20	1.59	215.58	59.97	27.62	3.99	0.96	1.65	0.11	0.03
	0.00	1.31	0.00	-13.93	31.06	5.83	1.44	2.04	0.17	0.04
32	1.00	2.13	1084.86	1149.77	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	1.76	869.57	727.89	14.07	1.66	0.33	0.66	0.05	0.01
	0.50	1.29	549.30	280.58	26.55	4.25	1.11	1.58	0.12	0.03
	0.20	1.02	324.89	101.25	33.35	5.24	1.33	2.13	0.15	0.04
	0.00	0.80	0.00	-18.57	37.89	7.05	1.92	2.64	0.21	0.05
64	1.00	1.64	1667.25	1736.23	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	1.41	1331.54	1105.49	18.72	1.92	0.22	0.88	0.05	0.01
	0.50	1.03	833.09	422.87	32.83	4.39	0.89	2.02	0.13	0.02
	0.20	0.85	498.63	147.82	38.66	6.09	1.55	2.70	0.19	0.04
	0.00	0.69	0.00	-24.40	45.49	8.42	2.10	3.57	0.26	0.06
128	1.00	1.57	2774.52	2860.34	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	1.44	2221.09	1844.18	34.97	8.42	3.03	1.85	0.25	0.09
	0.50	1.11	1390.18	705.12	57.02	20.49	7.75	4.73	0.67	0.22
	0.20	0.99	828.29	230.49	63.63	25.78	10.27	6.57	0.92	0.31
	0.00	0.78	0.00	-50.42	72.08	36.48	14.92	8.65	1.43	0.47

Table 5: Reconstruction with Saliency Map on Cora dataset using the GCN model.

Number of clusters	**Rate**	**Execution time (s)**	**Δnodes (%)**	**Δedges (%)**	**Affected explanations (%)**			**Affected nodes (%)**		
					0.2	0.5	0.7	0.2	0.5	0.7
1	1.00	16.96	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
16	1.00	2.89	720.35	774.97	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	2.49	576.00	498.22	10.78	1.14	0.41	0.46	0.03	0.01
	0.50	1.85	361.63	182.91	22.23	3.06	0.44	1.22	0.09	0.01
	0.20	1.56	218.28	60.21	27.92	4.36	1.40	1.66	0.13	0.04
	0.00	1.33	0.00	-13.93	31.28	6.13	1.51	2.05	0.18	0.04
32	1.00	2.02	1084.86	1149.77	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	1.66	871.05	729.23	15.14	2.18	0.52	0.71	0.06	0.01
	0.50	1.23	544.09	291.68	26.62	4.43	1.03	1.52	0.13	0.03
	0.20	0.99	323.38	96.23	32.75	5.58	1.26	2.09	0.16	0.04
	0.00	0.81	0.00	-18.57	37.92	7.42	2.03	2.62	0.22	0.06
64	1.00	1.55	1667.25	1736.23	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	1.33	1335.19	1109.95	17.32	1.81	0.26	0.76	0.05	0.01
	0.50	1.00	838.07	424.99	33.97	4.51	0.92	2.06	0.13	0.03
	0.20	0.83	498.71	139.39	39.88	6.61	1.48	2.78	0.20	0.04
	0.00	0.71	0.00	-24.40	45.64	8.79	2.18	3.58	0.27	0.06
128	1.00	1.52	2774.52	2860.34	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	1.37	2221.42	1827.06	34.42	9.60	2.88	1.82	0.29	0.08
	0.50	1.08	1385.89	710.21	56.65	20.75	8.60	4.74	0.70	0.25
	0.20	0.98	829.28	236.55	63.81	27.33	11.08	6.56	0.97	0.33
	0.00	0.79	0.00	-50.42	72.16	36.74	15.32	8.58	1.44	0.48

Table 6: Reconstruction with DeconvNet on the Cora dataset using the GAT model.

Number of clusters	**Rate**	**Execution time (s)**	**Δnodes (%)**	**Δedges (%)**	**Affected explanations (%)**			**Affected nodes (%)**		
					0.2	0.5	0.7	0.2	0.5	0.7
1	1.00	27.32	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
16	1.00	4.47	720.35	774.97	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	3.80	579.06	512.77	9.45	6.79	6.09	0.82	0.40	0.29
	0.50	2.79	364.62	191.82	23.23	16.65	14.55	2.12	0.96	0.67
	0.20	2.44	215.03	65.95	30.50	22.90	20.46	2.86	1.40	1.02
	0.00	2.06	0.00	-13.93	36.19	27.03	24.34	3.47	1.65	1.22
32	1.00	2.87	1084.86	1149.77	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	2.41	866.25	745.26	13.29	7.20	5.83	0.90	0.37	0.25
	0.50	1.76	541.54	285.13	30.80	23.19	20.49	2.92	1.41	1.03
	0.20	1.48	329.87	99.79	39.84	29.95	27.25	3.86	1.87	1.36
	0.00	1.17	0.00	-18.57	45.42	34.49	31.65	4.52	2.22	1.62
64	1.00	2.14	1667.25	1736.23	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	1.83	1334.01	1117.39	19.17	13.52	11.89	1.60	0.78	0.55
	0.50	1.38	831.54	418.85	38.11	28.10	25.00	3.40	1.61	1.18
	0.20	1.07	498.67	141.72	44.72	33.97	30.35	4.25	2.00	1.45
	0.00	0.85	0.00	-24.40	54.43	43.02	39.07	5.50	2.70	1.97
128	1.00	1.98	2774.52	2860.34	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	1.80	2226.29	1830.98	29.58	20.16	17.17	2.37	1.11	0.79
	0.50	1.31	1388.70	705.51	50.30	37.44	32.83	4.71	2.26	1.61
	0.20	1.10	842.87	233.93	61.63	51.22	47.23	6.48	3.28	2.43
	0.00	0.88	0.00	-50.42	70.35	60.71	56.20	7.38	3.94	2.96

Table 7: Reconstruction with GNNExplainer on the Cora dataset using the GAT model.

Number of clusters	**Rate**	**Execution time (s)**	**Δnodes (%)**	**Δedges (%)**	**Affected explanations (%)**			**Affected nodes (%)**		
					0.2	0.5	0.7	0.2	0.5	0.7
1	1.00	3086.39	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
16	1.00	280.26	323.08	318.78	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	249.97	260.38	211.03	45.46	45.46	45.46	10.45	10.45	10.45
	0.50	224.15	163.59	72.05	63.26	63.26	63.26	31.98	31.98	31.98
	0.20	212.70	97.82	26.73	65.84	65.84	65.84	34.57	34.57	34.57
	0.00	200.49	0.00	-13.93	69.76	69.76	69.76	44.25	44.25	44.25
32	1.00	156.02	433.71	416.12	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	137.44	346.90	267.73	53.10	53.10	53.10	16.11	16.11	16.11
	0.50	115.68	214.40	91.76	66.91	66.91	66.91	37.08	37.08	37.08
	0.20	108.06	130.72	24.93	71.34	71.34	71.34	48.76	48.76	48.76
	0.00	100.37	0.00	-18.57	75.41	75.41	75.41	57.89	57.89	57.89
64	1.00	92.54	623.12	583.50	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	78.11	498.71	365.97	59.31	59.31	59.31	20.64	20.64	20.64
	0.50	61.34	313.74	133.10	72.49	72.49	72.49	47.45	47.45	47.45
	0.20	54.83	183.60	39.58	75.92	75.92	75.92	57.75	57.75	57.75
	0.00	49.65	0.00	-24.40	79.10	79.10	79.10	67.83	67.83	67.83
128	1.00	62.84	1006.94	937.12	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	51.86	806.43	595.02	72.05	72.05	72.05	30.13	30.13	30.13
	0.50	37.06	504.73	223.66	82.31	82.31	82.31	55.72	55.72	55.72
	0.20	30.09	300.07	57.88	84.16	84.16	84.16	70.93	70.93	70.93
	0.00	25.84	0.00	-50.42	86.45	86.45	86.45	83.99	83.99	83.99

Table 8: Reconstruction with Guided Backprop on the Cora dataset using the GAT model.

Number of clusters	**Rate**	**Execution time (s)**	**Δnodes (%)**	**Δedges (%)**	**Affected explanations (%)**			**Affected nodes (%)**		
					0.2	0.5	0.7	0.2	0.5	0.7
1	1.00	27.72	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
16	1.00	4.38	720.35	774.97	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	3.69	577.36	492.12	9.05	5.98	5.10	0.76	0.34	0.23
	0.50	2.79	359.75	187.86	23.49	16.43	14.77	2.07	0.97	0.70
	0.20	2.41	218.02	58.83	31.50	23.49	21.86	3.02	1.49	1.11
	0.00	2.12	0.00	-13.93	35.82	26.99	24.37	3.44	1.65	1.23
32	1.00	2.91	1084.86	1149.77	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	2.44	868.43	749.43	15.07	9.64	8.35	1.19	0.52	0.38
	0.50	1.78	547.27	286.36	27.77	20.49	18.21	2.67	1.25	0.89
	0.20	1.49	324.48	94.30	34.27	25.00	21.79	3.17	1.51	1.07
	0.00	1.20	0.00	-18.57	45.05	34.75	31.68	4.49	2.22	1.62
64	1.00	2.14	1667.25	1736.23	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	1.85	1331.24	1088.97	18.57	12.59	11.37	1.60	0.74	0.55
	0.50	1.33	835.27	436.40	33.09	23.23	20.53	3.02	1.36	1.01
	0.20	1.12	501.85	147.59	40.32	29.73	26.66	3.84	1.78	1.29
	0.00	0.89	0.00	-24.40	54.17	42.98	38.88	5.46	2.69	1.97
128	1.00	1.99	2774.52	2860.34	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	1.78	2216.32	1840.83	23.67	15.44	12.70	1.93	0.85	0.59
	0.50	1.37	1395.27	722.85	49.89	38.44	34.71	4.93	2.37	1.70
	0.20	1.10	833.09	226.75	59.60	48.74	44.61	6.26	3.12	2.31
	0.00	0.89	0.00	-50.42	70.24	60.64	56.31	7.36	3.92	2.95

Table 9: Reconstruction with Saliency Map on the Cora dataset using the GAT model.

Number of clusters	**Rate**	**Execution time (s)**	**Δnodes (%)**	**Δedges (%)**	**Affected explanations (%)**			**Affected nodes (%)**		
					0.2	0.5	0.7	0.2	0.5	0.7
1	1.00	28.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
16	1.00	4.30	720.35	774.97	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	3.62	579.25	506.63	10.64	5.91	4.69	0.82	0.31	0.21
	0.50	2.83	360.78	188.27	20.57	13.59	11.56	1.85	0.77	0.50
	0.20	2.39	215.44	57.09	32.83	24.67	21.75	3.15	1.42	0.95
	0.00	2.06	0.00	-13.93	37.22	27.70	24.78	3.59	1.62	1.09
32	1.00	2.91	1084.86	1149.77	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	2.45	866.58	741.32	15.55	9.71	8.09	1.32	0.55	0.36
	0.50	1.78	542.73	276.85	27.95	18.13	15.66	2.38	0.98	0.68
	0.20	1.43	321.45	86.21	37.00	26.00	23.01	3.53	1.48	1.04
	0.00	1.18	0.00	-18.57	46.09	34.86	31.28	4.57	2.10	1.46
64	1.00	2.18	1667.25	1736.23	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	1.83	1333.68	1110.21	18.54	12.63	10.89	1.50	0.68	0.46
	0.50	1.33	831.65	416.88	35.52	24.74	21.01	3.23	1.37	0.94
	0.20	1.07	503.47	146.82	45.86	34.42	30.61	4.43	1.97	1.38
	0.00	0.84	0.00	-24.40	54.43	43.24	38.55	5.60	2.59	1.79
128	1.00	2.05	2774.52	2860.34	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	1.77	2217.87	1829.73	28.99	19.09	15.69	2.31	1.00	0.67
	0.50	1.31	1387.04	701.04	52.62	39.00	34.01	5.11	2.28	1.55
	0.20	1.12	831.35	230.73	63.81	50.41	45.57	6.47	3.10	2.16
	0.00	0.88	0.00	-50.42	71.05	60.45	55.61	7.59	3.87	2.79

Table 10: Reconstruction with DeconvNet on the CiteSeer dataset using the GCN model.

Number of clusters	**Rate**	**Execution time (s)**	**Δnodes (%)**	**Δedges (%)**	**Affected explanations (%)**			**Affected nodes (%)**		
					0.2	0.5	0.7	0.2	0.5	0.7
1	1.00	21.56	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
16	1.00	2.40	144.27	216.10	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	2.24	115.33	133.48	3.88	0.51	0.21	0.42	0.03	0.01
	0.50	1.93	72.86	54.57	7.69	1.20	0.36	0.97	0.08	0.02
	0.20	1.80	43.70	15.55	10.31	1.50	0.51	1.39	0.11	0.03
	0.00	1.67	0.00	-6.72	11.72	2.31	0.81	1.68	0.17	0.05
32	1.00	1.57	244.27	372.58	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	1.44	194.98	240.95	4.87	0.48	0.15	0.53	0.03	0.01
	0.50	1.25	119.81	89.81	10.46	1.26	0.24	1.40	0.09	0.02
	0.20	1.09	72.11	27.59	13.04	2.01	0.39	1.93	0.14	0.03
	0.00	0.96	0.00	-9.47	15.18	2.58	0.57	2.27	0.18	0.04
64	1.00	1.21	383.68	582.29	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	1.12	306.94	364.48	7.87	1.17	0.39	0.89	0.09	0.03
	0.50	0.95	195.94	142.82	13.20	1.92	0.54	1.86	0.14	0.04
	0.20	0.88	113.13	45.47	15.72	2.37	0.75	2.49	0.18	0.05
	0.00	0.73	0.00	-13.16	18.52	3.40	1.05	3.23	0.25	0.07
128	1.00	1.23	647.40	1012.32	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	1.32	520.47	649.30	14.10	3.07	0.90	1.70	0.21	0.06
	0.50	1.10	324.29	251.43	24.17	7.75	2.71	4.55	0.60	0.19
	0.20	0.99	192.61	73.95	28.16	10.37	3.52	6.45	0.89	0.25
	0.00	0.90	0.00	-29.09	32.37	14.43	5.50	8.40	1.34	0.40

Table 11: Reconstruction with GNNExplainer on the CiteSeer dataset using the GCN model.

Number of clusters	**Rate**	**Execution time (s)**	**Δnodes (%)**	**Δedges (%)**	**Affected explanations (%)**			**Affected nodes (%)**		
					0.2	0.5	0.7	0.2	0.5	0.7
1	1.00	2340.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
16	1.00	177.05	66.82	94.16	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	171.26	53.80	62.06	18.97	18.88	18.61	6.48	6.41	6.39
	0.50	159.65	33.00	22.50	25.82	25.64	25.16	18.25	17.82	17.75
	0.20	155.17	20.59	5.73	28.70	28.58	28.10	21.94	21.47	21.23
	0.00	151.34	0.00	-6.72	32.01	31.83	31.20	26.46	25.94	25.70
32	1.00	99.32	111.39	169.16	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	92.51	87.83	106.96	24.05	23.93	23.59	9.03	8.87	8.83
	0.50	84.39	54.40	37.96	31.14	30.84	30.33	23.48	23.06	22.95
	0.20	79.30	32.76	11.64	33.12	32.91	32.34	27.66	27.25	27.04
	0.00	75.84	0.00	-9.47	36.34	36.01	35.29	35.81	35.19	35.02
64	1.00	57.27	163.90	251.89	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	50.12	129.76	155.27	25.37	25.19	24.86	13.84	13.68	13.62
	0.50	43.70	82.21	61.25	33.81	33.54	33.03	29.09	28.61	28.40
	0.20	40.65	49.44	16.37	36.25	36.04	35.38	36.34	35.82	35.55
	0.00	37.81	0.00	-13.16	39.59	39.34	38.53	44.81	44.07	43.82
128	1.00	37.46	285.18	454.28	0.03	0.00	0.00	0.00	0.00	0.00
	0.70	32.45	228.25	282.25	31.44	31.35	30.63	19.44	19.11	18.82
	0.50	25.62	142.11	101.74	42.59	42.20	41.24	40.94	39.94	39.72
	0.20	22.38	84.25	21.95	45.63	45.27	43.97	52.64	51.61	51.16
	0.00	19.48	0.00	-29.09	48.90	48.39	46.92	62.36	61.02	60.58

Table 12: Reconstruction with Guided Backprop on the CiteSeer dataset using the GCN model.

Number of clusters	**Rate**	**Execution time (s)**	**Δnodes (%)**	**Δedges (%)**	**Affected explanations (%)**			**Affected nodes (%)**		
					0.2	0.5	0.7	0.2	0.5	0.7
1	1.00	21.74	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
16	1.00	2.41	144.27	216.10	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	2.26	115.87	142.42	4.39	0.57	0.24	0.47	0.04	0.02
	0.50	1.94	72.35	51.34	8.21	1.14	0.33	1.03	0.08	0.02
	0.20	1.81	42.41	16.83	9.53	1.68	0.57	1.38	0.12	0.04
	0.00	1.69	0.00	-6.72	11.72	2.31	0.81	1.68	0.17	0.05
32	1.00	1.58	244.27	372.58	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	1.44	195.16	240.14	4.78	0.60	0.12	0.52	0.04	0.01
	0.50	1.22	123.47	95.47	10.37	1.32	0.18	1.40	0.09	0.01
	0.20	1.15	71.30	27.37	13.20	1.95	0.48	1.88	0.14	0.03
	0.00	0.96	0.00	-9.47	15.18	2.58	0.57	2.27	0.18	0.04
64	1.00	1.18	383.68	582.29	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	1.16	305.98	374.43	7.42	0.63	0.09	0.78	0.04	0.01
	0.50	0.95	191.88	144.68	12.44	1.80	0.51	1.82	0.14	0.03
	0.20	0.86	116.65	48.55	15.96	2.71	0.78	2.57	0.20	0.05
	0.00	0.77	0.00	-13.16	18.52	3.40	1.05	3.23	0.25	0.07
128	1.00	1.24	647.40	1012.32	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	1.28	518.52	649.45	14.97	3.58	1.20	1.93	0.26	0.08
	0.50	1.12	325.04	247.10	24.56	8.00	2.92	4.63	0.61	0.20
	0.20	0.97	192.82	72.69	28.13	10.43	3.31	6.40	0.85	0.23
	0.00	0.85	0.00	-29.09	32.37	14.43	5.50	8.41	1.34	0.40

Table 13: Reconstruction with Saliency Map on the CiteSeer dataset using the GCN model.

Number of clusters	**Rate**	**Execution time (s)**	**Δnodes (%)**	**Δedges (%)**	**Affected explanations (%)**			**Affected nodes (%)**		
					0.2	0.5	0.7	0.2	0.5	0.7
1	1.00	21.96	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
16	1.00	2.39	144.27	216.10	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	2.18	115.57	137.85	3.61	0.36	0.12	0.37	0.03	0.01
	0.50	1.96	70.33	48.57	8.51	1.23	0.36	1.07	0.08	0.02
	0.20	1.80	41.33	14.96	10.31	1.62	0.51	1.41	0.11	0.03
	0.00	1.64	0.00	-6.72	11.75	2.31	0.81	1.68	0.17	0.05
32	1.00	1.56	244.27	372.58	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	1.47	195.40	244.90	4.90	0.51	0.06	0.55	0.04	0.00
	0.50	1.19	124.17	92.60	9.95	1.41	0.33	1.39	0.10	0.02
	0.20	1.11	72.47	28.10	12.92	1.98	0.39	1.87	0.14	0.03
	0.00	0.95	0.00	-9.47	15.00	2.55	0.57	2.26	0.18	0.04
64	1.00	1.17	383.68	582.29	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	1.15	306.40	366.48	6.55	0.75	0.09	0.73	0.05	0.01
	0.50	0.96	193.00	141.83	13.89	1.86	0.45	1.96	0.14	0.03
	0.20	0.83	112.32	43.63	16.20	2.68	0.66	2.62	0.19	0.04
	0.00	0.77	0.00	-13.16	18.27	3.49	0.93	3.22	0.26	0.06
128	1.00	1.28	647.40	1012.32	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	1.26	513.53	628.51	13.29	3.28	1.23	1.81	0.23	0.08
	0.50	1.12	326.21	250.59	24.05	7.81	2.46	4.49	0.60	0.17
	0.20	0.97	194.17	76.71	27.83	10.31	3.46	6.13	0.82	0.24
	0.00	0.84	0.00	-29.09	32.25	14.58	5.65	8.29	1.36	0.41

Table 14: Reconstruction with DeconvNet on the CiteSeer dataset using the GAT model.

Number of clusters	**Rate**	**Execution time (s)**	**Δnodes (%)**	**Δedges (%)**	**Affected explanations (%)**			**Affected nodes (%)**		
					0.2	0.5	0.7	0.2	0.5	0.7
1	1.00	35.61	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
16	1.00	3.84	144.27	216.10	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	3.55	115.09	132.07	4.78	3.49	2.98	1.14	0.56	0.41
	0.50	3.24	70.24	50.92	10.52	7.60	6.70	2.32	1.16	0.86
	0.20	3.05	44.88	16.72	13.07	9.41	8.66	3.09	1.53	1.14
	0.00	2.93	0.00	-6.72	15.06	10.79	9.59	3.46	1.72	1.27
32	1.00	2.34	244.27	372.58	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	2.04	194.77	232.29	4.57	2.86	2.01	0.92	0.38	0.25
	0.50	1.80	121.07	88.14	13.41	9.65	8.36	3.00	1.46	1.07
	0.20	1.62	71.63	28.89	15.03	11.18	9.68	3.54	1.77	1.25
	0.00	1.43	0.00	-9.47	19.36	14.73	12.71	4.60	2.37	1.73
64	1.00	1.63	383.68	582.29	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	1.51	306.76	368.91	6.07	3.85	3.19	1.25	0.57	0.40
	0.50	1.24	194.35	145.65	13.83	9.32	8.30	2.94	1.43	1.05
	0.20	1.17	115.72	48.95	18.43	13.44	11.69	4.24	2.12	1.57
	0.00	0.98	0.00	-13.16	23.47	17.67	15.63	5.46	2.81	2.12
128	1.00	1.50	647.40	1012.32	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	1.54	520.17	663.22	11.39	7.06	5.95	2.28	0.98	0.70
	0.50	1.25	323.53	247.80	22.45	16.11	13.89	4.88	2.38	1.71
	0.20	1.12	195.58	85.52	26.99	20.08	18.18	6.25	3.26	2.41
	0.00	0.99	0.00	-29.09	33.36	26.54	24.02	7.99	4.30	3.23

Table 15: Reconstruction with GNNExplainer on the CiteSeer dataset using the GAT model.

Number of clusters	**Rate**	**Execution time (s)**	**Δnodes (%)**	**Δedges (%)**	**Affected explanations (%)**			**Affected nodes (%)**		
					0.2	0.5	0.7	0.2	0.5	0.7
1	1.00	3816.54	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
16	1.00	282.86	66.82	94.16	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	270.56	53.56	61.45	18.61	8.48	6.73	6.96	2.51	1.26
	0.50	260.07	33.72	22.30	26.03	14.73	12.77	20.07	7.52	2.61
	0.20	253.08	20.05	4.77	29.79	18.27	16.38	27.03	10.31	3.87
	0.00	248.29	0.00	-6.72	32.01	21.43	19.36	32.59	12.14	4.63
32	1.00	156.86	111.39	169.16	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	147.42	88.10	105.93	23.08	11.33	9.62	10.92	4.10	1.76
	0.50	135.74	57.29	40.73	29.76	18.27	16.41	23.40	8.59	3.76
	0.20	129.44	33.24	10.39	32.64	21.40	19.39	33.42	11.21	4.93
	0.00	124.75	0.00	-9.47	36.31	26.27	24.38	42.42	15.08	6.56
64	1.00	87.09	163.90	251.89	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	79.55	132.25	160.65	24.05	14.01	12.23	21.02	9.61	2.25
	0.50	70.73	83.68	61.34	33.93	21.97	19.96	32.05	10.83	5.26
	0.20	65.75	51.58	19.75	36.16	26.33	24.05	43.27	15.32	6.57
	0.00	62.05	0.00	-13.16	39.59	31.20	29.10	52.36	18.31	8.61
128	1.00	55.93	285.18	454.28	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	48.55	228.94	294.09	31.05	18.52	16.20	21.32	8.99	3.63
	0.50	40.34	141.27	101.85	40.58	28.94	26.57	44.55	15.97	8.54
	0.20	36.00	86.83	28.91	45.60	36.25	33.81	57.53	19.78	11.32
	0.00	31.93	0.00	-29.09	48.90	42.29	39.98	66.60	22.85	14.44

Table 16: Reconstruction with Guided Backprop on the CiteSeer dataset using the GAT model.

Number of clusters	**Rate**	**Execution time (s)**	**Δnodes (%)**	**Δedges (%)**	**Affected explanations (%)**			**Affected nodes (%)**		
					0.2	0.5	0.7	0.2	0.5	0.7
1	1.00	36.03	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
16	1.00	3.53	144.27	216.10	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	3.30	114.07	135.90	3.40	2.22	2.04	0.80	0.39	0.28
	0.50	3.03	72.77	51.41	11.51	8.54	7.54	2.65	1.32	0.96
	0.20	2.97	43.37	17.60	9.53	5.98	5.02	1.92	0.87	0.63
	0.00	2.86	0.00	-6.72	15.06	10.76	9.56	3.46	1.72	1.27
32	1.00	2.39	244.27	372.58	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	2.20	196.60	248.51	3.94	2.49	1.95	0.68	0.33	0.24
	0.50	1.78	122.15	90.75	13.80	10.82	9.92	3.41	1.72	1.29
	0.20	1.66	74.84	32.56	12.14	8.72	7.21	2.84	1.37	0.95
	0.00	1.45	0.00	-9.47	19.33	14.70	12.71	4.59	2.37	1.72
64	1.00	1.59	383.68	582.29	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	1.49	307.91	380.89	7.42	4.81	4.18	1.64	0.76	0.58
	0.50	1.29	192.34	149.34	14.52	10.07	8.72	3.14	1.48	1.13
	0.20	1.13	116.35	50.53	19.63	14.25	12.89	4.71	2.35	1.74
	0.00	1.00	0.00	-13.16	23.44	17.64	15.63	5.46	2.81	2.13
128	1.00	1.51	647.40	1012.32	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	1.52	519.09	663.03	8.75	4.51	3.88	1.58	0.64	0.46
	0.50	1.27	323.87	247.43	21.52	15.12	13.32	4.62	2.29	1.70
	0.20	1.16	192.73	76.49	27.41	20.47	17.70	6.15	3.10	2.30
	0.00	0.97	0.00	-29.09	33.27	26.51	23.96	7.98	4.30	3.22

Table 17: Reconstruction with Saliency Map on the CiteSeer dataset using the GAT model.

Number of clusters	**Rate**	**Execution time (s)**	**Δnodes (%)**	**Δedges (%)**	**Affected explanations (%)**			**Affected nodes (%)**		
					0.2	0.5	0.7	0.2	0.5	0.7
1	1.00	35.65	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
16	1.00	3.57	144.27	216.10	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	3.28	116.95	141.10	2.74	1.41	1.08	0.49	0.17	0.11
	0.50	2.95	71.51	48.68	9.86	6.55	5.80	2.17	1.00	0.73
	0.20	2.77	44.63	18.26	13.23	9.35	8.24	2.93	1.43	1.03
	0.00	2.62	0.00	-6.72	15.21	10.88	9.65	3.45	1.69	1.26
32	1.00	2.24	244.27	372.58	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	2.04	194.89	227.68	6.31	4.90	4.42	1.30	0.67	0.51
	0.50	1.81	123.93	93.48	10.55	7.42	6.46	2.43	1.13	0.80
	0.20	1.62	73.37	29.31	14.91	10.70	9.44	3.33	1.63	1.17
	0.00	1.43	0.00	-9.47	19.36	14.82	13.07	4.52	2.31	1.67
64	1.00	1.62	383.68	582.29	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	1.48	307.94	366.98	7.12	5.05	4.57	1.65	0.76	0.56
	0.50	1.23	191.16	142.07	14.64	10.04	8.81	3.23	1.51	1.08
	0.20	1.15	116.71	50.70	18.12	12.89	11.57	4.13	2.01	1.47
	0.00	0.96	0.00	-13.16	23.69	17.73	15.87	5.45	2.76	2.07
128	1.00	1.56	647.40	1012.32	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	1.53	517.64	649.36	11.21	7.18	6.04	2.29	1.00	0.74
	0.50	1.25	322.03	243.06	22.90	16.53	14.34	5.03	2.40	1.77
	0.20	1.13	193.45	77.33	27.65	20.95	18.88	6.48	3.24	2.37
	0.00	0.95	0.00	-29.09	33.36	26.69	24.17	7.94	4.21	3.16

Table 18: Reconstruction with DeconvNet on the PubMed dataset using the GCN model.

Number of clusters	**Rate**	**Execution time (s)**	**Δnodes (%)**	**Δedges (%)**	**Affected explanations (%)**			**Affected nodes (%)**		
					0.2	0.5	0.7	0.2	0.5	0.7
1	1.00	283.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
16	1.00	133.09	875.45	1020.18	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	96.07	701.05	654.14	9.91	0.47	0.07	0.35	0.01	0.00
	0.50	53.63	436.46	252.70	17.68	1.20	0.19	0.81	0.02	0.00
	0.20	34.68	261.85	82.38	21.03	1.87	0.23	1.11	0.04	0.00
	0.00	17.80	0.00	-16.70	24.19	2.27	0.34	1.26	0.04	0.01

Table 19: Reconstruction with Guided Backprop on the PubMed dataset using the GCN model.

Number of clusters	Rate	Execution time (s)	Δnodes (%)	Δedges (%)	Affected explanations (%)			Affected nodes (%)		
					0.2	0.5	0.7	0.2	0.5	0.7
1	1.00	283.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
16	1.00	133.50	875.45	1020.18	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	96.65	700.08	651.77	9.63	0.55	0.08	0.34	0.01	0.00
	0.50	54.00	437.61	251.60	17.19	1.35	0.16	0.79	0.03	0.00
	0.20	34.75	261.46	84.25	21.56	1.83	0.25	1.12	0.03	0.00
	0.00	18.26	0.00	-16.70	24.20	2.28	0.34	1.26	0.04	0.01

Table 20: Reconstruction with Saliency Map on the PubMed dataset using the GCN model

Number of clusters	Rate	Execution time (s)	Δnodes (%)	Δedges (%)	Affected explanations (%)			Affected nodes (%)		
					0.2	0.5	0.7	0.2	0.5	0.7
1	1.00	294.67	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
16	1.00	132.81	875.45	1020.18	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	95.81	699.60	650.67	9.60	0.50	0.07	0.32	0.01	0.00
	0.50	53.65	438.81	248.93	16.91	1.32	0.20	0.76	0.02	0.00
	0.20	34.38	261.90	83.51	21.51	1.97	0.30	1.09	0.04	0.01
	0.00	17.79	0.00	-16.70	24.05	2.42	0.39	1.22	0.05	0.01

Table 21: Reconstruction with DeconvNet on the PubMed dataset using the GAT model.

Number of clusters	Rate	Execution time (s)	Δnodes (%)	Δedges (%)	Affected explanations (%)			Affected nodes (%)		
					0.2	0.5	0.7	0.2	0.5	0.7
1	1.00	403.32	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
16	1.00	182.39	875.45	1020.18	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	132.23	700.79	648.32	16.07	6.44	4.13	0.68	0.18	0.09
	0.50	75.30	438.18	251.91	29.48	13.71	9.59	1.65	0.43	0.23
	0.20	48.17	262.44	82.84	35.72	18.18	13.00	2.22	0.60	0.33
	0.00	25.03	0.00	-16.70	41.61	21.70	16.18	2.78	0.76	0.42

Table 22: Reconstruction with Guided Backprop on the PubMed dataset using the GAT model.

Number of clusters	**Rate**	**Execution time (s)**	**Δnodes (%)**	**Δedges (%)**	**Affected explanations (%)**			**Affected nodes (%)**		
					0.2	0.5	0.7	0.2	0.5	0.7
1	1.00	399.74	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
16	1.00	182.89	875.45	1020.18	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	132.46	700.33	651.08	16.07	6.33	4.12	0.72	0.18	0.10
	0.50	75.51	437.71	252.07	29.30	13.67	9.71	1.68	0.44	0.24
	0.20	48.27	264.23	85.06	34.61	17.41	12.47	2.13	0.58	0.31
	0.00	25.64	0.00	-16.70	41.61	21.70	16.18	2.78	0.76	0.42

Table 23: Reconstruction with Saliency Map on the PubMed dataset using the GAT model.

Number of clusters	**Rate**	**Execution time (s)**	**Δnodes (%)**	**Δedges (%)**	**Affected explanations (%)**			**Affected nodes (%)**		
					0.2	0.5	0.7	0.2	0.5	0.7
1	1.00	401.21	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
16	1.00	182.07	875.45	1020.18	0.00	0.00	0.00	0.00	0.00	0.00
	0.70	132.18	701.16	653.18	15.97	6.31	4.22	0.71	0.19	0.11
	0.50	75.33	440.30	250.29	30.67	14.18	9.93	1.71	0.45	0.24
	0.20	48.14	263.62	83.85	35.62	17.55	12.81	2.19	0.59	0.33
	0.00	25.10	0.00	-16.70	41.65	21.61	16.13	2.78	0.76	0.42